Master Documentation _Generated 2025-10-29 06:08_

Total files: 231 **Total size:** 43.5 MB

File Types

{'py': 42, 'txt': 5, 'md': 5, 'pdf': 22, 'noext': 3, 'pkl': 7, 'ipynb': 22, 'yml': 4, 'json': 1, 'jpg': 32, 'xlsx': 4, 'png': 48, 'csv': 28, 'joblib': 5, 'db': 1, 'toml': 1, 'gitkeep': 1}

Repository Tree

.devcontainer - `.devcontainer/devcontainer.json` (1.5 KB) ### .git - `.git/FETCH\_HEAD` (139.0 B) - `.git/HEAD` (21.0 B) - `.git/config` (428.0 B) - `.git/config.worktree` (83.0 B) - `.git/description` (73.0 B) - `.git/hooks/applypatch-msg.sample` (478.0 B) - `.git/hooks/commit-msg.sample` (896.0 B) - `.git/hooks/fsmonitor-watchman.sample` (4.6 KB) - `.git/hooks/post-update.sample` (189.0 B) - `.git/hooks/pre-applypatch.sample` (424.0 B) - `.git/hooks/pre-commit.sample` (1.6 KB) - `.git/hooks/pre-merge-commit.sample` (416.0 B) - `.git/hooks/pre-push.sample` (1.3 KB) - `.git/hooks/pre-rebase.sample` (4.8 KB) - `.git/hooks/pre-receive.sample` (544.0 B) - `.git/hooks/prepare-commit-msg.sample` (1.5 KB) - `.git/hooks/push-to-checkout.sample` (2.7 KB) - `.git/hooks/sendemail-validate.sample` (2.3 KB) - `.git/hooks/update.sample` (3.6 KB) - `.git/index` (25.2 KB) - `.git/info/exclude` (240.0 B) - `.git/logs/HEAD` (218.0 B) - `.git/logs/refs/heads/main` (227.0 B) - `.git/logs/refs/remotes/origin/main` (326.0 B) - `.git/objects/pack/pack-da3920c81d234310e14768c3f` (8.1 KB) - `.git/objects/pack/pack-da3920c81d234310e14768c3fa4a91c032c1abbf.pack` (20.3 MB) - `.git/objects/pack/pack-da3920c81d234310e14768c3fa4a91c032c1abbf.rev` (1.1 KB) - `.git/refs/heads/main` (41.0 B) - `.git/refs/remotes/origin/main` (41.0 B) - `.git/shallow` (41.0 B) ### .github - `.github/workflows/autodoc.yml` (5.1 KB) - `.github/workflows/autodoc\_universe.yml` (6.1 KB) - `.github/workflows/gen-tree.yml` (949.0 B) - `.github/workflows/generate-pdf.yml` (1.6 KB) ### backups - `backups/sample.txt` (32.0 B) ### cb_dashboard_artifacts - `cb\_dashboard\_artifacts/README.md` (5.7 KB) - `cb\_dashboard\_artifacts/data/benchmarking\_data\_fakergen.xlsx` (12.0 KB) - `cb\_dashboard\_artifacts/data/employee\_compensation\_fakergen.xlsx` (1.6 MB) - `cb\_dashboard\_artifacts/exports/CB\_User\_Guide.pdf` (3.4 KB) - `cb\_dashboard\_artifacts/exports/Cb\_Report\_consolidated.pdf` (564.5 KB) - `cb\_dashboard\_artifacts/images/app\_layout/1-App-Landing-Page.jpg` (192.2 KB) - `cb\_dashboard\_artifacts/images/app\_layout/2-Step-1-Download-Templates-Guides.jp` (170.6 KB) - `cb\_dashboard\_artifacts/images/app\_layout/3-Guide.jpg` (246.1 KB) - `cb\_dashboard\_artifacts/images/app\_layout/4-Upload-Data.jpg` (130.0 KB) - `cb\_dashboard\_artifacts/images/app\_layout/5-Set-Job-Order.jpg` (188.9 KB) - `cb\_dashboard\_artifacts/images/app\_layout/6-Appy-Order.jpg` (93.8 KB) - `cb\_dashboard\_artifacts/images/app\_layout/7-Session-Persistence.jpg` (206.7 KB) - `cb\_dashboard\_artifacts/images/app\_layout/8-PDF-Downloads-Section.jpg` (189.4 KB) - `cb\_dashboard\_artifacts/images/app\_layout/9-Image-Downloads-Section.jpg` (206.6 KB) - `cb\_dashboard\_artifacts/images/chatbot/Chatbot-1.jpg` (77.3 KB) - `cb\_dashboard\_artifacts/images/chatbot/Chatbot-2.jpg`

(121.5 KB) - cb_dashboard_artifacts/images/chatbot/Chatbot-3.jpg
(77.5 KB) - cb_dashboard_artifacts/images/chatbot/Chatbot-4.jpg
(114.9 KB) - cb_dashboard_artifacts/images/chatbot/Chatbot-5.jpg
(127.9 KB) - cb_dashboard_artifacts/images/chatbot/Chatbot-6.jpg
(160.0 KB) - cb_dashboard_artifacts/images/metrics/Metric-1-Graph.jpg
(94.9 KB) - cb_dashboard_artifacts/images/metrics/Metric-1-Table-A.jpg
(83.9 KB) - cb_dashboard_artifacts/images/metrics/Metric-1-Table-B.jpg
(443.6 KB) - cb_dashboard_artifacts/images/metrics/Metric-2-Graph.jpg
(95.5 KB) - cb_dashboard_artifacts/images/metrics/Metric-2-Table.jpg
(118.0 KB) - cb_dashboard_artifacts/images/metrics/Metric-3-Graph.jpg
(48.4 KB) - cb_dashboard_artifacts/images/metrics/Metric-3-Table.jpg
(127.3 KB) - cb_dashboard_artifacts/images/metrics/Metric-4-Graph.jpg
(110.9 KB) - cb_dashboard_artifacts/images/metrics/Metric-4-Table.jpg
(114.1 KB) - cb_dashboard_artifacts/images/metrics/Metric-5-Graph.jpg
(93.6 KB) - cb_dashboard_artifacts/images/metrics/Metric-5-Table.jpg
(140.6 KB) - cb_dashboard_artifacts/images/metrics/Metric-6-Graph.jpg
(104.7 KB) - cb_dashboard_artifacts/images/metrics/Metric-6-Table.jpg
(153.1 KB) - cb_dashboard_artifacts/images/metrics/Metric-7-Graph.jpg
(106.0 KB) - cb_dashboard_artifacts/images/metrics/Metric-7-Table.jpg
(164.8 KB) ### charts - charts/filetypes.png (39.2 KB) ### data -
data/Attrition_SQL_Predictions.csv (235.6 KB) - data/Comp_Analytics_Processed.csv
(25.3 KB) - data/bonus_by_department.csv (742.0 B) - data/cleaned_hr_data.csv
(248.4 KB) - data/comp_ctc_by_department.csv (276.0 B) - data/comp_median_ctc.csv
(127.0 B) - data/day4_bonus_vs_perf.csv (127.0 B) - data/day4_ctc_by_dept.csv
(245.0 B) - data/day4_gender_paygap.csv (85.0 B) - data/day4_joblevel_counts.csv
(158.0 B) - data/day5_ctc_quartiles.csv (447.0 B) - data/day5_gender_gap_by_level.csv
(719.0 B) - data/employee_attrition.csv (225.5 KB) - data/employee_compensation_sample.csv
(16.5 KB) - data/hr_dataset.db (220.0 KB) - data/logistic_top_features.csv
(1.8 KB) - data/messy_hr_data.csv (247.6 KB) - data/model_comparison_results.csv
(191.0 B) - data/processed_hr_data.csv (225.5 KB) - data/rf_top_features.csv
(1.8 KB) - data/shap_feature_importance_full.csv (1.8 KB) -
data/xgb_top_features.csv (1.4 KB) ### faker_data - faker_data/FakerGen_CB.ipynb
(10.2 KB) - faker_data/benchmarking_data_fakergen.xlsx (12.0 KB)
- faker_data/employee_compensation_fakergen.xlsx (1.6 MB) -
faker_data/employee_master.csv (8.2 MB) ### helper_scripts -
helper_scripts/hr_db.py (547.0 B) - helper_scripts/sql_utils.py
(926.0 B) ### images - images/Attrition_by_dept.jpg (328.2 KB) -
images/attrition_by_age.jpg (116.5 KB) - images/attrition_by_jobrole.png
(264.7 KB) - images/bonus_by_department.png (69.2 KB) - images/comp_bonus_dist.png
(25.9 KB) - images/comp_ctc_by_dept.png (23.8 KB) - images/comp_ctc_by_joblevel.png
(15.1 KB) - images/comp_gender_gap.png (24.4 KB) - images/comp_median_ctc.png
(19.0 KB) - images/confusion_matrix.png (38.4 KB) - images/confusion_matrix_xgb.png
(54.5 KB) - images/day4_bonus_vs_perf.png (38.1 KB) - images/day4_ctc_by_dept.png
(37.6 KB) - images/day4_gender_paygap.png (20.3 KB) - images/day4_joblevel_counts.png
(42.9 KB) - images/day5_bonus_kde.png (49.9 KB) - images/day5_ctc_quartiles.png
(37.6 KB) - images/day5_gender_gap_by_level.png (38.0 KB) -

images/department_attrition.png (94.1 KB) - images/donut_attrition.png
(83.1 KB) - images/missing_values_after.png (379.1 KB) - images/missing_values_before.png
(379.7 KB) - images/missing_values_collage.png (404.6 KB) -
images/model_comparision.png (75.4 KB) - images/model_comparison_barplot.png
(107.2 KB) - images/probability_distribution.png (94.6 KB) -
images/roc_curve_comparison.png (170.2 KB) - images/screenshot1.png
(23.0 B) - images/shap_dependence_plot.png (119.1 KB) - images/shap_dependence_plot_v2.png
(77.6 KB) - images/shap_feature_importance.png (325.4 KB) -
images/shap_feature_importance_top15.png (142.4 KB) - images/shap_local_bar_plot.png
(90.6 KB) - images/shap_summary_plot.png (479.6 KB) - images/shap_summary_plot_V2.png
(79.4 KB) - images/shap_waterfall_employee0.png (67.9 KB) -
images/shap_waterfall_plot.png (96.4 KB) - images/streamlit_db_1.png
(197.2 KB) - images/streamlit_db_2.png (62.4 KB) - images/streamlit_db_3.png
(146.6 KB) - images/streamlit_db_4.png (118.5 KB) - images/top_features.png
(72.8 KB) - images/top_features_post_enhancement.png (160.2 KB) -
images/top_features_tuned.png (165.5 KB) - images/top_features_xgb.png
(155.2 KB) - images/top_features_xgb_heatmap.png (271.9 KB)
models - models/Attrition_AdvancedModels.ipynb (281.6 KB) -
models/logistic_attrition_model.pkl (1.2 KB) - models/logistic_attrition_model_tuned.pkl
(2.6 KB) - models/placeholder (28.0 B) - models/random_forest_attrition_model.pkl
(4.9 MB) - models/random_forest_tuned.pkl (1.4 MB) - models/scaler.pkl
(3.1 KB) - models/trained_columns.pkl (1.0 KB) - models/xgboost_attrition_model.pkl
(368.7 KB) ### models_joblib - models_joblib/logistic_attrition_model.joblib
(1.2 KB) - models_joblib/random_forest_attrition_model.joblib
(4.6 MB) - models_joblib/scaler.joblib (2.8 KB) - models_joblib/trained_columns.joblib
(996.0 B) - models_joblib/xgboost_attrition_model.joblib (493.6 KB)
notebooks - notebooks/ Attrition_PredictiveModel.ipynb (109.6
KB) - notebooks/.gitkeep (1.0 B) - notebooks/Attrition_AdvancedModels.ipynb
(281.6 KB) - notebooks/Attrition_ModelComparision.ipynb (130.8
KB) - notebooks/Attrition_ModelExplainability.ipynb (865.7 KB)
- notebooks/Attrition_PredictiveModel_V2.ipynb (150.0 KB) -
notebooks/Attrition_PredictiveModel_V3.ipynb (94.2 KB) - notebooks/Attrition_SHAP_Explainab
(731.1 KB) - notebooks/Attrition_SQL_Integration-Git.ipynb
(123.1 KB) - notebooks/Attrition_Streamlit_Dashboard.ipynb
(114.0 B) - notebooks/Compensation_Analytics_V1.ipynb (185.2
KB) - notebooks/Compensation_Analytics_V2.ipynb (215.0 KB) -
notebooks/Compensation_Analytics_V3.ipynb (213.8 KB) - notebooks/Compensation_Analytics_V4.
(195.1 KB) - notebooks/Day4-AttritionRiskAnalyzer.ipynb (4.8 KB)
- notebooks/Day4-AttritionRiskAnalyzer_v2.0.ipynb (176.9 KB) -
notebooks/baby_steps/Day1-Hello-Amlan.ipynb (983.0 B) - notebooks/baby_steps/Day2-Basics.ipyn
(2.5 KB) - notebooks/baby_steps/Day3-DataTypesAndControl.ipynb (3.6
KB) - notebooks/baby_steps.gitkeep (1.0 B) ### people_analytics -
people_analytics/.streamlit/config.toml (314.0 B) - people_analytics/HR_DataForge.ipynb
(130.5 KB) - people_analytics/README.md (1.0 B) - people_analytics/app.py
(6.5 KB) - people_analytics/faker_data/Attrition_Data.csv (10.6
KB) - people_analytics/faker_data/Compensation_Benchmark.csv

(151.0 B) - people_analytics/faker_data/Compensation_Internal.csv
(8.9 KB) - people_analytics/faker_data/Engagement_Survey.csv
(7.2 KB) - people_analytics/faker_data/Performance_Data.csv
(10.0 KB) - people_analytics/faker_data/Workforce_Data.csv
(10.2 KB) - people_analytics/modules/attrition_module.py (8.2
KB) - people_analytics/modules/compensation_module.py (8.8 KB)
- people_analytics/modules/consolidated_module.py (4.1 KB) -
people_analytics/modules/consolidated_module_diagnostic.py
(2.9 KB) - people_analytics/modules/engagement_module.py (7.8
KB) - people_analytics/modules/performance_module.py (6.7 KB) -
people_analytics/modules/workforce_module.py (7.2 KB) - people_analytics/pages/attrition.py
(244.0 B) - people_analytics/pages/compensation.py (256.0 B) -
people_analytics/pages/consolidated.py (9.5 KB) - people_analytics/pages/engagement.py
(248.0 B) - people_analytics/pages/performance.py (252.0 B) -
people_analytics/pages/workforce.py (244.0 B) - people_analytics/requirements.txt
(326.0 B) - people_analytics/templates/test.txt (1.0 B) - people_analytics/utils/__init__.py
(535.0 B) - people_analytics/utils/chart_saver.py (4.7 KB) -
people_analytics/utils/fix_helper.py (1.3 KB) - people_analytics/utils/fusion_report_buil
(6.2 KB) - people_analytics/utils/pdf_auto_exporter.py (6.7 KB)
- people_analytics/utils/pdf_explainer_builder.py (14.1 KB) -
people_analytics/utils/pdf_helper.py (7.2 KB) - people_analytics/utils/persistence_chatbot
(5.5 KB) - people_analytics/utils/template_helper.py (1.2 KB) -
people_analytics/utils/ui_styling.py (2.4 KB) - people_analytics/utils/uploader_helper.py
(1.2 KB) - people_analytics/utils_consolidated/__init__.py
(691.0 B) - people_analytics/utils_consolidated/chart_consolidated_saver.py
(1.4 KB) - people_analytics/utils_consolidated/constants.py (858.0
B) - people_analytics/utils_consolidated/deck_state_tracker.py
(2.3 KB) - people_analytics/utils_consolidated/insights_helper.py
(515.0 B) - people_analytics/utils_consolidated/pdf_consolidated_helper.py
(8.9 KB) - people_analytics/utils_consolidated/pdf_helper_consolidated.py
(826.0 B) - people_analytics/utils_consolidated/pdf_merger.py (9.2
KB) - people_analytics/utils_consolidated/toc_helper.py (633.0 B)
- people_analytics/utils_consolidated/uploader_consolidated_helper.py
(1.5 KB) ### reports - reports/Attrition_AdvancedModels_aa41c1310f907082a92a9e51b94010b5.pdf
(278.5 KB) - reports/Attrition_ModelComparision_009086358233572ffb338cf8f4ee6ab7.pdf
(176.5 KB) - reports/Attrition_ModelExplainability_52d61660976d1481ab051617939779a2.pdf
(441.2 KB) - reports/Attrition_Project_Summary.pdf (199.3 KB) -
reports/Attrition_SHAP_Explainability_V1_883d0c6ff33ddb40c5459b894a884b62.pdf
(308.6 KB) - reports/Attrition_SQL_Integration-Git_5bb7a8192e09a762c9dbe6088839adf4.pdf
(151.2 KB) - reports/Attrition_SQL_Integration_967dced0f92426fc0889ff773fe5ac24.pdf
(178.0 KB) - reports/Compensation_Analytics_V1_07f883af597f1eb48eed37f6749c53f2.pdf
(167.6 KB) - reports/Compensation_Analytics_V1_0d3ab6f265e56379d7bfa3752feec2a5.pdf
(167.4 KB) - reports/Compensation_Analytics_V1_b3320c085ce3523bb04ed8c2f02b2cd6.pdf
(33.6 KB) - reports/Compensation_Analytics_V2_96c5a8ec0ee8e61b721ddf37c74b4f4a.pdf
(184.9 KB) - reports/Compensation_Analytics_V3_045981c9a6b93fc4e15735327218b694.pdf
(206.3 KB) - reports/Compensation_Analytics_V4_2f677fec7d2d33e580d73985b739f3b6.pdf

(199.3 KB) - reports/Day4-AttritionRiskAnalyzer_v2.0_058b58b9bf5e96ac0c1bdaac1e7abdcc.pdf
(188.8 KB) - reports/HR_Data_Cleaning_Utility_V1_260622ce.pdf
(238.8 KB) - reports/HR_Data_Cleaning_Utility_V1_6501dfa8.pdf
(238.8 KB) - reports/HR_Data_Cleaning_Utility_V1_86b38b95.pdf
(218.4 KB) - reports/HR_Data_Cleaning_Utility_V1_87664d4cc3eefa67f9fe549d33ef7c57.pdf
(218.4 KB) ### scripts - scripts/export_pdf.py (3.4 KB) - scripts/gen_tree.py
(1.8 KB) ### sidequests - sidequests/.gitkeep (1.0 B) - sidequests/HR_Data_Cleaning_Utility_V1
(257.0 KB) ### universe_charts - universe_charts/Personal_Apps.png
(13.7 KB) - universe_charts/SQL-Fundamentals-Lab.png (13.1 KB) - universe_charts/hr-tech-portfolio.png (32.9 KB) - universe_charts/python-fundamentals-lab.png
(15.2 KB)

Code & Notebook Overview

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/generate_autodoc.py

```
import pathlib, os, datetime, subprocess, collections, nbformatROOT = pathlib.Path('.').reso
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/cb_dashboard.py

```
\# =====================================\# cb\_dashboard.py - Compensation \& Benefits Dash
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/app.py

```
\# app.py -- HR Attrition Prediction Dashboard (Final Polished Version)\# Save this file as
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/models/Attrition_AdvancedModels

```
\#  Project 5: Advanced Attrition Models (Tuned Random Forest + XGBoost)

In this notebook, we benchmark and enhance predictive models for employee attrition.
- Start with tuned **Random Forest**
- Introduce **XGBoost** (gradient boosting)
- Compare performance vs Logistic Regression

**Goal:**
Improve predictive accuracy while balancing interpretability and business storytelling.
\# -----------------------
\# Imports
\# -----------------------
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import os
import joblib

from sklearn.model\_sele
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/scripts/export_pdf.py

```
import osimport subprocessimport hashlibfrom pathlib import Pathfrom nbconvert import HTMLEx
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/scripts/gen_tree.py

```
import osfrom pathlib import Pathdef generate\_tree(start\_path=".", prefix="", max\_items=3
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/faker_data/FakerGen_CB.ipynb

```
<a href="https://colab.research.google.com/github/AMBOT-pixel96/hr-tech-portfolio/blob/main/
\# -------------------------------
\# Step 0 - Clone hr-tech-portfolio (with PAT)
\# -------------------------------
import os
from getpass import getpass

\# Ask for your GitHub Personal Access Token
token = getpass(" Enter your GitHub token: ")

\# Clean any previous clone
os.system("rm -rf hr-tech-portfolio")

\# Clone with token authentication
os.system(f"git clone
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/helper_scripts/sql_utils.py

```
import sqlite3, pandas as pd, osDB\_PATH = os.path.abspath("hr\_dataset.db")def run\_query(s
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/helper_scripts/hr_db.py

```
import pandas as pd, sqlite3, osDB\_PATH = os.path.abspath("hr\_dataset.db")CSV\_PATH = os.p
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/HR_DataForge.i

```
<a href="https://colab.research.google.com/github/AMBOT-pixel96/hr-tech-portfolio/blob/main/
!pip install faker pandas numpy
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/app.py

```
\# ============================================\# app.py - People Analytics Dashboard (v3.9
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/utils/fusion_r

```
\# ============================================\# utils/fusion\_report\_builder.py - v2.3 |
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/utils/ui_styli

```
\# ============================================\# utils/ui\_styling.py - v1.0 | Unified Side
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/utils/pdf_expl

```
\# ============================================\# utils/pdf\_explainer\_builder.py - v5.8 |
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/utils/chart_sa

```
\# ============================================\# utils/chart\_saver.py - v4.4 | ColorSafe E
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/utils/pdf_auto

\# ==\# utils/pdf_auto_exporter.py - v3.7 | File

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/utils/persisten

\# ==\# utils/persistence_chatbot.py - v2.6 | Res

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/utils/fix_help

\# ==\# utils/fix_helper.py - v3.0.3 | Final Sync

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/utils/template\

```
import streamlit as stimport pandas as pd\# --- Unified Styling for All Download Buttons ---
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/utils/uploader\

\# ==\# utils_consolidated/uploader_consolidated

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/utils/__init\

"""utils package initializer-------------------------Ensures core utility modules load clean

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/utils/pdf_help

\# ==\# utils/pdf_helper.py - v5.2 | Executive St

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/pages/compensat

```
import streamlit as stst.set\_page\_config(page\_title="Compensation Analytics", layout="wid
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/pages/workforce

```
import streamlit as stst.set\_page\_config(page\_title="Workforce Analytics", layout="wide")
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/pages/engagemen

```
import streamlit as stst.set\_page\_config(page\_title="Engagement Analytics", layout="wide"
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/pages/performan

```
import streamlit as stst.set\_page\_config(page\_title="Performance Analytics", layout="wide
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/pages/consolida

\# ==\# pages/consolidated.py - v6.8 | Auto Job De

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/pages/attrition

```
import streamlit as stst.set\_page\_config(page\_title="Attrition Analytics", layout="wide")
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/utils_consolid

\# ==\# utils_consolidated/deck_state_tracker.p

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/utils_consolid

\# ==\# utils_consolidated/chart_consolidated_s

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/utils_consolid

\# utils_consolidated/insights_helper.py"""Small helper that consolidates insights lists

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/utils_consolid

\# utils_consolidated/constants.py\# Theme + layout constants for consolidated PDF builder

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/utils_consolid

\# ==\# utils_consolidated/pdf_helper_consolid

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/utils_consolid

\# ==\# utils_consolidated/pdf_merger.py - v7.5

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/utils_consolid

\# utils_consolidated/uploader_consolidated_helper.py\# v1.3 - uploader that is mobile-f

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/utils_consolid

\# utils_consolidated/toc_helper.py"""Simple TOC helper: builds a table-friendly TOC arra

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/utils_consolid

\# ==\# utils_consolidated/pdf_consolidated_he

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/utils_consolid

"""utils_consolidated package initializer-----------------------------------Ensures all

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/modules/consoli

\# ==\# modules/consolidated_module.py - v7.0 |

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/modules/workfo

\# ==\# modules/workforce_module.py - v3.0.1 | E

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/modules/engage

\# ==\# modules/engagement_module.py - v3.0.1 |

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/modules/consoli

\# ==\# consolidated_module_diagnostic.py - Min

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/modules/attriti

\# ==\# modules/attrition_module.py - v3.2 | Con

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/modules/perfor

\# ==\# modules/performance_module.py - v3.0.1 |

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/people_analytics/modules/compen

\# ==\# modules/compensation_module.py - v3.0.1

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Attrition_SQL_Integr

```
\# Project 8: SQL + ML Integration

**Objective:**
Combine **SQL querying power** with **Machine Learning models** to analyze attrition risk.
This project demonstrates how HR teams can query their employee database directly and run pr

**Why It Matters:**
- HR data often lives in databases (HRIS, payroll systems).
- Analysts should be able to run queries and pipe results into ML models.
- This integration makes predictive attrition analytics more practical in enterprise context
\#\#  SQL + ML Integration

Thi
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Day4-AttritionRiskAnaly

```
\# Attrition Risk Analyzer (v2.0)

\#\# Objective
Analyze IBM HR Attrition dataset to identify attrition patterns and create a simple attritic
\# Import required libraries
import pandas as pd
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Attrition_PredictiveMo

```
\# Predictive Model - Attrition (Logistic Regression)\_V3 - Fine Tuning
\# -----------------------
\# Imports
\# -----------------------
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import os

from sklearn.model\_selection import train\_test\_split, cross\_val\_score, GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.linear\_model import LogisticRegression
from sklearn.metrics import (
    accuracy\_score, classification\_report, confusion\_matrix, roc\_auc\_score
)
import joblib

sns.set(style="whitegrid")
pd.set\_option('display.max\_c
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Attrition_ModelCompari

```
\#  Project 4: Attrition Model Comparison (Logistic vs Random Forest)
```

```
In this notebook, we extend the attrition prediction by comparing **Logistic Regression** (I

**Goal:**
Check if tree-based models improve predictive performance and what new insights they provide
\# -----------------------
\# Imports
\# -----------------------
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import os

from sklearn.model\_selection import train\_test\_split
from sklearn.preprocessing import Standa
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Compensation_Analytics

```
<a href="https://colab.research.google.com/github/AMBOT-pixel96/hr-tech-portfolio/blob/main/
\# -------------------------------
\# Step 0 - Clone Repo with PAT Auth
\# -------------------------------

import os
from getpass import getpass

\# If repo already exists in Colab, remove it (fresh start)
!rm -rf hr-tech-portfolio

\# Ask for PAT token securely
token = getpass("Enter your GitHub PAT: ")

\# Clone using HTTPS + token auth
!git clone https:/
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Attrition_AdvancedMode

```
\#  Project 5: Advanced Attrition Models (Tuned Random Forest + XGBoost)

In this notebook, we benchmark and enhance predictive models for employee attrition.
- Start with tuned **Random Forest**
- Introduce **XGBoost** (gradient boosting)
- Compare performance vs Logistic Regression

**Goal:**
Improve predictive accuracy while balancing interpretability and business storytelling.
\# -----------------------
```

```
\# Imports
\# ------------------------
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import os
import joblib

from sklearn.model\_sele
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Attrition_PredictiveMo

```
\# Predictive Model - Attrition (Logistic Regression)
**Objective:** Train a logictic regression model to predict employee attrition (Yes/No) usin
\# Imports
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
sns.set(style="whitegrid")

from sklearn.model\_selection import train\_test\_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear\_model import LogisticRegression
from sklearn.metrics import accuracy\_score, classification\_report, confusion\_matrix, roc\
import os

\# Display settings
pd.set\_option('displ
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Day4-AttritionRiskAnaly

```
import pandas as pd
df= pd.read\_csv("WA\_Fn-UseC\_-HR-Employee-Attrition.csv")
print("Shape of Data:",df.shape) \#Explore Dataset
print(df.head())
print(df['Attrition'].value\_counts()) \#Quick Summary of Attrition
\#Group by Job Role
job\_attrition=df.groupby(['JobRole','Attrition']).size()
job\_attrition=job\_attrition.unstack().fillna(0)
print(job\_attrition)
\#Simple Attrition Risk Flag
df['AttritionRisk']=df['Attrition'].apply(lambda x:1 if x=="Yes" else 0)
\#Save Processed File
df.to\_csv("processed\_hr\_data.csv",index=False)
print("Attrition Risk Analyzer v0.1 ready  ")
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Compensation_Analytics

```
<a href="https://colab.research.google.com/github/AMBOT-pixel96/hr-tech-portfolio/blob/main/
from getpass import getpass

\# GitHub authentication
token = getpass(" Enter your GitHub token: ")

repo\_url = f"https://AMBOT-pixel96:\{token\}@github.com/AMBOT-pixel96/hr-tech-portfolio.git

!git clone \{repo\_url\}
\%cd hr-tech-portfolio
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/ Attrition_PredictiveModel.ipynb

```
\# Predictive Model - Attrition (Logistic Regression)
**Objective:** Train a logictic regression model to predict employee attrition (Yes/No) usin
\# Imports
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
sns.set(style="whitegrid")

from sklearn.model\_selection import train\_test\_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear\_model import LogisticRegression
from sklearn.metrics import accuracy\_score, classification\_report, confusion\_matrix, roc\
import joblib
import sqlite3

\# displ
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Attrition_ModelExplain

```
\#  Project 6: Model Explainability with SHAP

In this notebook, we apply **SHAP (SHapley Additive exPlanations)** to interpret machine lea

**Goals:**
- Understand which features drive attrition globally across employees
- Explain individual predictions for specific employees
- Provide visuals that HR leaders can use to trust model decisions
\# 1  Imports
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Attrition_Streamlit_

> *(Could not preview: Notebook does not appear to be JSON: '# Save training columns for alignment\n...)* ### /home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/

```
\# Project 9: Attrition Explainability with SHAP

**Objective:**
Enhance the HR Attrition prediction models by adding explainability using SHAP (SHapley Addi
This notebook will:
1. Load saved models from Project 7 (Deployment).
2. Apply SHAP to explain feature contributions.
3. Export explainability visuals for reports and dashboards.

**Artifacts generated:**
- SHAP summary plots
- SHAP bar plots
- SHAP force plots (employee-level explanations)
- Stored outputs in `/reports/09\_explainability/`

import pandas as pd
import shap
import joblib
import os
fr
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Compensation_Analytics

```
<a href="https://colab.research.google.com/github/AMBOT-pixel96/hr-tech-portfolio/blob/main/
\# Project 10: Compensation Analytics (v1.0)

**Objective:**
Analyze synthetic HR Compensation dataset to extract key insights: pay distribution, gender

**Business Context:**
Compensation \& Benefits (C\&B) analytics helps HR leaders answer:
- Are we paying fairly across levels, genders, departments?
- How are bon
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/Compensation_Analytics

```
<a href="https://colab.research.google.com/github/AMBOT-pixel96/hr-tech-portfolio/blob/main/

\# Clone repo + move into it
from getpass import getpass
token = getpass(" Enter your GitHub token: ")

repo\_url = f"https://AMBOT-pixel96:\{token\}@github.com/AMBOT-pixel96/hr-tech-portfolio.git
!git clone \{repo\_url\}
\%cd hr-tech-portfolio

\# Import libraries
import pandas as pd
```

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/baby_steps/Day2-Basics

```
\#1.Variables
name="Amlan"
age="29"
print("My name is",name," and I am",age," years old")
\#2.Simple Math
salary=1404000 \#Yearly INR
bonus=salary*.12
total=round(salary+bonus)
print("Total Annual Pay:",total)
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/baby_steps/Day1-Hello-

```
print("Hello Amlan - HR Tech Transformation  ")
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/notebooks/baby_steps/Day3-DataTy

```
\#1.Dictionaries
employee=\{
    "name":"Amlan",
    "role":"HR Tech",
    "salary":1404000,
    "skills":["PowerBI","Oracle Fusion","Python"]
\}
print(employee["name"],"earns",employee["salary"],"as a",employee["role"],"professional with
\#2.If-Else
attrition\_risk =0.7
if attrition\_risk > 0.5:
    print("High Risk - Action Needed")
else:
    print("Low Risk - All Good")
```

/home/runner/work/hr-tech-portfolio/hr-tech-portfolio/sidequests/HR_Data_Cleaning_Ut

```
\#  HR Data Cleaning Utilities (v1.0)

This notebook demonstrates how to **simulate messy HR data** and then build a cleaning pipel
Data cleaning is a critical step in People Analytics - poor quality data = misleading insigh
\# Import libraries
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import random
```